

Chaoran Chen

Curriculum Vitae

Department of Computer Science and Engineering
College of Engineering
University of Notre Dame
Notre Dame, IN 46556 USA

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Research Interests

Human-centered AI governance; AI agents and human oversight; privacy auditing; usable privacy and security; human-AI interaction.

Research Profile

My research studies how AI systems reshape user agency, privacy, and oversight across AI-mediated interactions, agentic workflows, and model training pipelines. I design and evaluate sociotechnical mechanisms for contextual privacy scaffolding, behavioral auditing, and human oversight of LLM-powered agents. My recent work focuses on auditing unauthorized use of protected data in RL fine-tuning, supporting privacy decisions in LLM applications, and evaluating human oversight failures in GUI agents.

Education

Ph.D. in Computer Science and Engineering 2022 – 2026
University of Notre Dame, *Notre Dame, IN*
Department of Computer Science and Engineering
Advisors: Toby Jia-Jun Li and Yanfang Ye

M.S. in Educational Technology and Applied Learning Sciences 2021 – 2022
Carnegie Mellon University, *Pittsburgh, PA*
Human-Computer Interaction Institute, School of Computer Science

B.Eng. in Industrial Design 2015 – 2020
Tongji University, *Shanghai, China*
College of Design and Innovation

Academic Appointments

Incoming Postdoctoral Fellow Sep. 2026 –
Johns Hopkins University, *Baltimore, MD, USA*
Whiting School of Engineering
Advisor: Yaxing Yao

Research Experience

Applied Scientist Intern May 2026 - Jul. 2026
Amazon, Seattle, WA, USA

Hosts: Vy Nguyen and Dr. Ziji Zhang

Research focus: robustness and evaluation of tool-using conversational agents under uncertainty.

Student Researcher

Jul. 2025 - Apr. 2026

Google, Sunnyvale, CA, USA

Hosts: Dr. Dayu Yuan and Dr. Peter Kairouz

Research focus: privacy auditing for RL fine-tuned LLMs, including behavioral canaries for detecting unauthorized use of protected retrieved contexts in post-training.

Research Intern

Oct. 2024 - Dec. 2024

Max Planck Institute for Security and Privacy, Bochum, Germany

Host: Dr. Abraham Mhaidli

Research focus: literature synthesis and research agenda development on privacy risks, human oversight, and security challenges in LLM-powered GUI agents.

Selected Fellowships, Programs, Honors, Awards, and Grants

International Programme on AI Evaluation: Capabilities & Safety, Inaugural Cohort, 2026

Advanced Researcher Award, University of Notre Dame, 2026

Travel Grant, Symposium on Usable Privacy and Security (SOUPS), 2025

Third Prize, Tech Ethics Hackathon (\$1,000), Notre Dame–IBM Technology Ethics Lab, 2024

Gary Marsden Travel Award, ACM SIGCHI, 2022

Best Implementation Award (\$2,000), MobileHCI 2022

Merit Scholarship (\$7,000), Carnegie Mellon University

Preprints

- [P.5] **Behavioral Canaries: Auditing Private Retrieved Context Usage in RL Fine-Tuning**
Chaoran Chen, Dayu Yuan, Peter Kairouz
arXiv preprint arXiv:2604.22191, 2026.

- [P.4] **Comparing Human Oversight Strategies for Computer-Use Agents**
Chaoran Chen, Zhiping Zhang, Zeya Chen, Eryue Xu, Yinuo Yang, Ibrahim Khalilov, Simret A Gebreegziabher, Yanfang Ye, Ziang Xiao, Yaxing Yao, Tianshi Li, Toby Jia-Jun Li
arXiv preprint arXiv:2604.04918, 2026.

- [P.3] **How Coding Agents Fail Their Users: A Large-Scale Analysis of Developer-Agent Misalignment in 20,574 Real-World Sessions**
Ningzhi Tang, **Chaoran Chen**, Gelei Xu, Yiyu Shi, Yu Huang, Collin McMillan, Tao Dong, and Toby Jia-Jun Li
arXiv preprint arXiv:2605.29442, 2026.

- [P.2] **Humans' ALMANAC: A Human Collaboration Dataset of Action-Level Mental Model Annotations for Agent Collaboration**
Jiaju Chen, Yuxuan Lu, Jiayi Su, **Chaoran Chen**, Songlin Xiao, Zheng Zhang, Yun Wang, Yunyao Li, Jian Zhao, Tongshuang Wu, Toby Jia-Jun Li, Dakuo Wang, and Bingsheng Yao
arXiv preprint arXiv:2606.06388, 2026.

- [P.1] **NARRA-Gym for Evaluating Interactive Narrative Agents**
Yue Huang, Yuchen Ma, Jiayi Ye, Wenjie Wang, Zipeng Ling, Xingjian Hu, Yuexing Hao, Zichen Chen, Zhangchen Xu, Yunhong He, Zhengqing Yuan, Yujun Zhou, Kehan Guo, **Chaoran Chen**,

Toby Jia-Jun Li, Stefan Feuerriegel, and Xiangliang Zhang
arXiv preprint arXiv:2605.08503, 2026.

Journal Papers

- [J.1] **The Obvious Invisible Threat: LLM-Powered GUI Agents’ Vulnerability to Fine-Print Injections**
Chaoran Chen, Zhiping Zhang, Bingcan Guo, Shang Ma, Ibrahim Khalilov, Simret Gebreegziabher, Yanfang Ye, Ziang Xiao, Yaxing Yao, Tianshi Li, and Toby Jia-Jun Li
ACM Transactions on AI Security and Privacy (TAISAP), 2026

Conference Papers

** indicates equal contribution*

- [C.11] **Programming by Chat: A Large-Scale Behavioral Analysis of 11,579 Real-World AI-Assisted IDE Sessions**
Ningzhi Tang*, Chaoran Chen*, Zian Fang, Gelei Xu, Maria Dhakal, Yiyu Shi, Collin McMillan, Yu Huang, and Toby Jia-Jun Li
Accepted to the 41st IEEE/ACM International Conference on Automated Software Engineering (ASE 2026)
- [C.10] **Stayin' Aligned Over Time: Towards Longitudinal Human-LLM Alignment via Contextual Reflection and Privacy-Preserving Behavioral Data**
Simret Araya Gebreegziabher, Allison E Sproul, Yinuo Yang, Chaoran Chen, Diego Gómez-Zarà, and Toby Jia-Jun Li
Conditionally accepted to the 39th Annual ACM Symposium on User Interface Software and Technology (UIST 2026)
- [C.9] **Dark Patterns Meet GUI Agents: LLM Agent Susceptibility to Manipulative Interfaces and the Role of Human Oversight**
Jingyu Tang*, Chaoran Chen*, Jiawen Li, Zhiping Zhang, Bingcan Guo, Ibrahim Khalilov, Simret Araya Gebreegziabher, Bingsheng Yao, Dakuo Wang, Yanfang Ye, Tianshi Li, Ziang Xiao, Yaxing Yao, and Toby Jia-Jun Li
Proceedings of the ACM Conference on Human Factors in Computing Systems (CHI 2026)
- [C.8] **Through the Lens of Human-Human Collaboration: A Configurable Research Platform for Exploring Human-Agent Collaboration**
Bingsheng Yao, Jiaju Chen, Chaoran Chen, April Wang, Toby Jia-jun Li, and Dakuo Wang
Proceedings of the ACM Conference on Human Factors in Computing Systems (CHI 2026)
- [C.7] **My Favorite Streamer is an LLM: Discovering, Bonding, and Co-Creating in AI VTuber Fandom**
Jiayi Ye, Chaoran Chen, Yue Huang, Yanfang Ye, Toby Jia-Jun Li and Xiangliang Zhang
Proceedings of the ACM Conference on Human Factors in Computing Systems (CHI 2026)
Best Paper Honorable Mention Award
- [C.6] **The Behavioral Fabric of LLM-Powered GUI Agents: Human Values and Interaction Outcomes**
Simret Araya Gebreegziabher, Yukun Yang, Charles Chiang, Hojun Yoo, Chaoran Chen, Hyo Jin Do, Zahra Ashktorab, Werner Geyer, Diego Gómez-Zarà, and Toby Jia-Jun Li

- [C.5] **Towards a Design Guideline for RPA Evaluation: A Survey of Large Language Model-Based Role-Playing Agents**
Chaoran Chen, Bingsheng Yao, Ruishi Zou, Wenyue Hua, Weimin Lyu, Toby Jia-Jun Li, and Dakuo Wang
Findings of the 63rd Annual Meeting of the Association for Computational Linguistics (ACL Findings 2025)
- [C.4] **Careful About What App Promotion Ads Recommend! Detecting and Explaining Malware Promotion via App Promotion Graph**
Shang Ma*, **Chaoran Chen***, Shao Yang, Shifu Hou, Toby Jia-Jun Li, Xusheng Xiao, Tao Xie, and Yanfang Ye
The Network and Distributed System Security Symposium (NDSS 2025)
- [C.3] **Why Am I Seeing This: Democratizing End User Auditing for Online Content Recommendations**
Chaoran Chen, Leyang Li, Luke Cao, Yanfang Ye, Tianshi Li, Yaxing Yao, and Toby Jia-Jun Li
Proceedings of the 38th Annual ACM Symposium on User Interface Software and Technology (UIST 2025)
- [C.2] **CLEAR: Towards Contextual LLM-Empowered Privacy Policy Analysis and Risk Generation for Large Language Model Applications**
Chaoran Chen, Daodao Zhou, Yanfang Ye, Toby Jia-jun Li, and Yaxing Yao
Proceedings of the 30th ACM Conference on Intelligent User Interfaces (IUI 2025)
- [C.1] **An Empathy-Based Sandbox Approach to Bridge the Privacy Gap among Attitudes, Goals, Knowledge, and Behaviors**
Chaoran Chen, Weijun Li, Wenxin Song, Yanfang Ye, Yaxing Yao, and Toby Jia-Jun Li
Proceedings of the ACM Conference on Human Factors in Computing Systems (CHI 2024)

Workshop Papers

- [W.6] **Beyond Task Completion: A Process-Oriented Framework for Evaluating Preference and Value Alignments in Computer-Using Agents**
Yukun Yang, Simret Araya Gebreegziabher, Hojun Yoo, Charles Chiang, **Chaoran Chen**, Annalisa Szymanski, Hyo Jin Do, Zahra Ashktorab, Werner Geyer, Diego Gómez-Zarà, and Toby Jia-Jun Li
IUI 2026 Workshop on Trustworthy Computer-Using Generalist Agents for Intelligent User Interfaces
- [W.5] **Iterative Contextual Consent: AI-enabled Data Privacy Contracts**
Daniel D Slate, **Chaoran Chen**, Yaxing Yao, Toby Jia-Jun Li
CCS 2025 Workshop on Human-Centered AI Privacy and Security
- [W.4] **Beyond Permissions: Investigating Mobile Personalization with Simulated Personas**
Ibrahim Khalilov, **Chaoran Chen**, Ziang Xiao, Tianshi Li, Toby Jia-Jun Li, Yaxing Yao
CCS 2025 Workshop on Human-Centered AI Privacy and Security
- [W.3] **Toward a Human-centered Evaluation Framework for Trustworthy LLM-powered GUI Agents**
Chaoran Chen, Zhiping Zhang, Ibrahim Khalilov, Bingcan Guo, Simret A. Gebreegziabher, Yanfang Ye, Ziang Xiao, Yaxing Yao, Tianshi Li, and Toby Jia-Jun Li

CHI 2025 Workshop on Human-Centered Evaluation and Auditing of Language Models (HEAL)

[W.2] **Evaluating the LLM Agents for Simulating Humanoid Behavior**
Chaoran Chen, Bingsheng Yao, Yanfang Ye, Dakuo Wang, and Toby Jia-Jun Li
CHI 2024 Workshop on Human-Centered Evaluation and Auditing of Language Models (HEAL)

[W.1] **Prompt Learning Unlocked for App Promotion in the Wild**
Zhongyu Ouyang, Shifu Hou, Shang Ma, **Chaoran Chen**, Chunhui Zhang, Toby Jia-Jun Li,
Xusheng Xiao, Chuxu Zhang, and Yanfang Ye
NeurIPS 2023 Workshop on New Frontiers in Graph Learning (GLFrontiers)

Lightly-Reviewed Publications and Extended Abstracts

[L.2] **EvalAgent: Interactive Comparative Evaluation of Computer-Using GUI Agents**
Yukun Yang, Simret Araya Gebreegziabher, Hojun Yoo, Charles Chiang, **Chaoran Chen**, Annalisa Szymanski, Hyo Jin Do, Zahra Ashktorab, Werner Geyer, Diego Gómez-Zarà, and Toby Jia-Jun Li
ACM Conference on Intelligent User Interfaces (IUI 2026) Demos Track

[L.1] **PriviSense: A Frida-Based Framework for Multi-Sensor Spoofing on Android**
Ibrahim Khalilov, **Chaoran Chen**, Ziang Xiao, Tianshi Li, Toby Jia-Jun Li, Yaxing Yao
2026 IEEE/ACM 48th International Conference on Software Engineering (ICSE 2026): Demo Track

Teaching Experience

Teaching Assistant, CSE 40567: Computer Security Spring 2025
Department of Computer Science and Engineering, University of Notre Dame

Teaching Assistant, CSE 40625: Machine Learning Fall 2023
Department of Computer Science and Engineering, University of Notre Dame

Teaching Assistant, 05-738: Evidence-Based Educational Design Fall 2022
Human-Computer Interaction Institute, Carnegie Mellon University

Teaching Assistant, 05-440/05-640: Interaction Techniques (IxD) Spring 2022
Human-Computer Interaction Institute, Carnegie Mellon University

Academic Service

Organizer / Program Committee Member

Human-Agent Collaboration Workshop at CHI 2026; HAIPS Workshop at CCS 2025

Conference Reviewer

CHI 2023–2026; CSCW 2025; IUI 2025–2026; UIST 2026; ICLR Trustworthy AI Workshop 2026

Received special recognitions for outstanding reviews at CHI 2024 LBW, CSCW 2025, IUI 2025–2026, CHI 2026, and UIST 2026.

Journal Reviewer

ACM TOPS; ACM TAISAP; ACM AI Letters; IHCS; PACM EICS